The University of Texas Rio Grande Valley

Wireshark Lab 1:

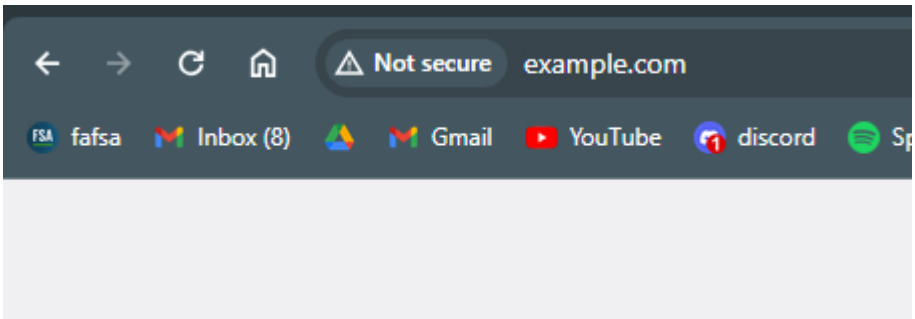
Introduction to Wireshark

The University of Texas
Rio Grande Valley

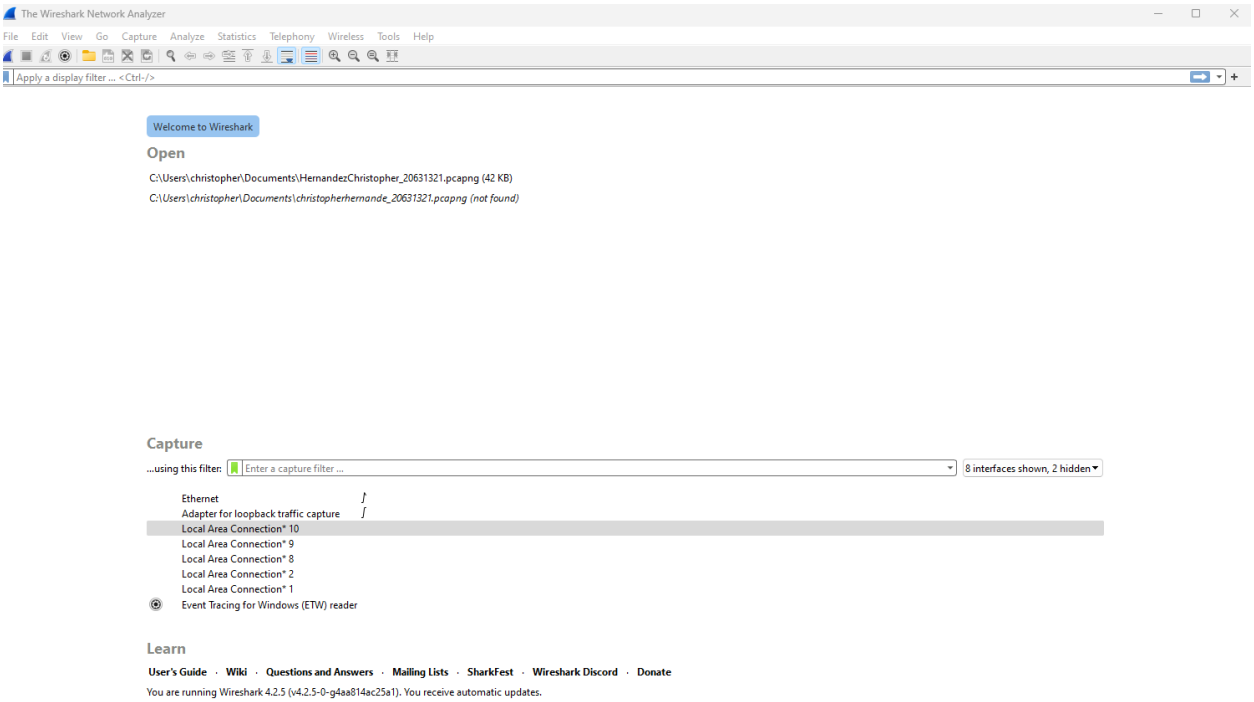
1. Start your Web Browser.

Type the name of a ***university or website*** (make sure it is http not https. There are somehttp sites for testing)to capture traffic, but do not hit return yet. I chose www.tsc.edu. Every student should select their own.

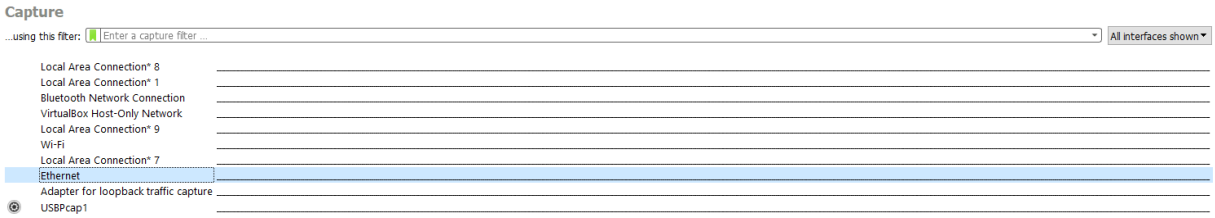
To make sure you are getting http traffic, make sure you type the name of the university as http://www.tsc.edu:80 << the http:// and the :80 make sure you will get some http traffic on the 80 port even if the http is not supported. For example, many will get that the http location moved to a new https location.



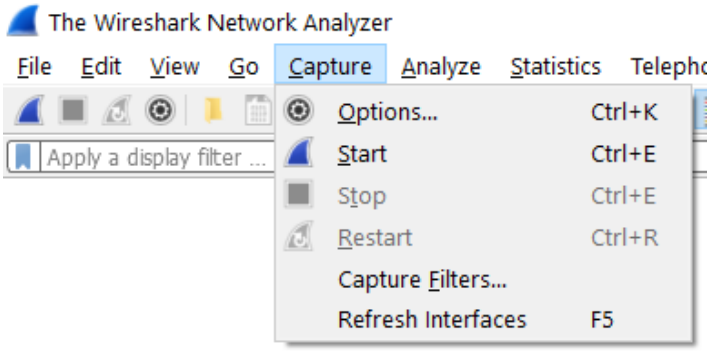
2. Start your Wireshark



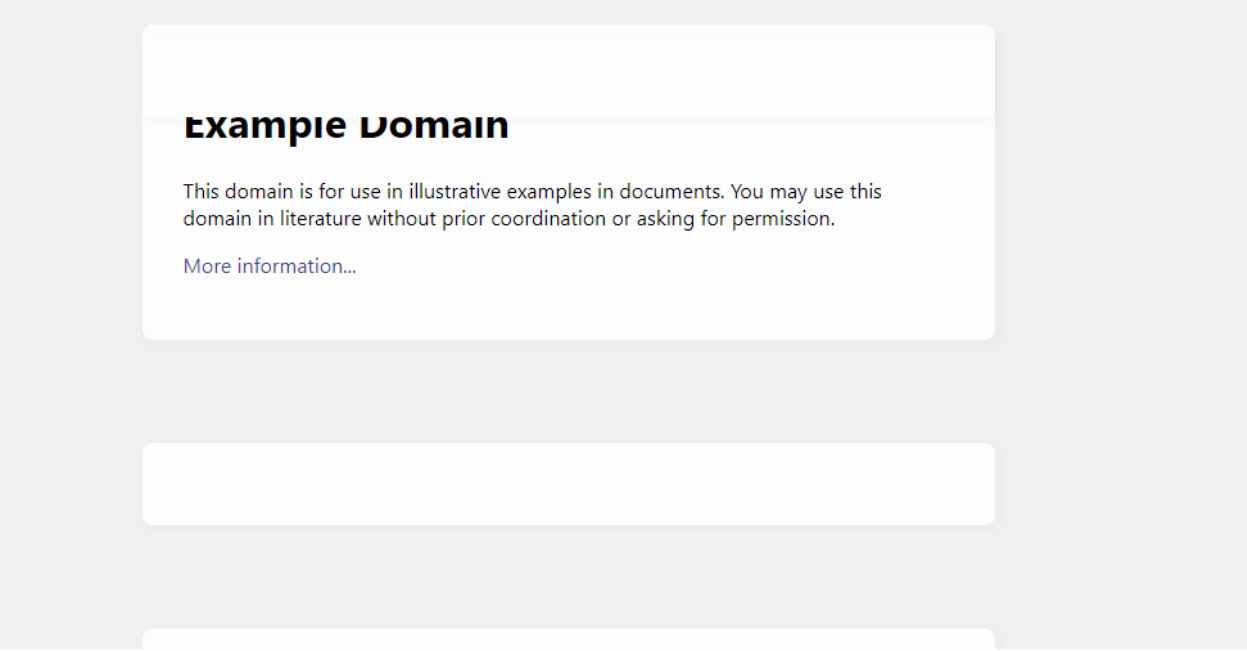
3. Select the interface that you want to capture traffic for



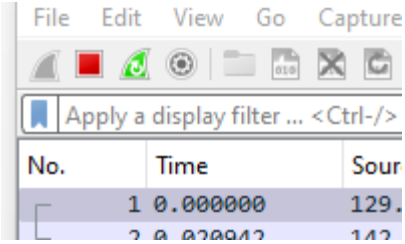
4. Select Capture and hit start. Quickly switch to the browser and hit enter



I use a testing website

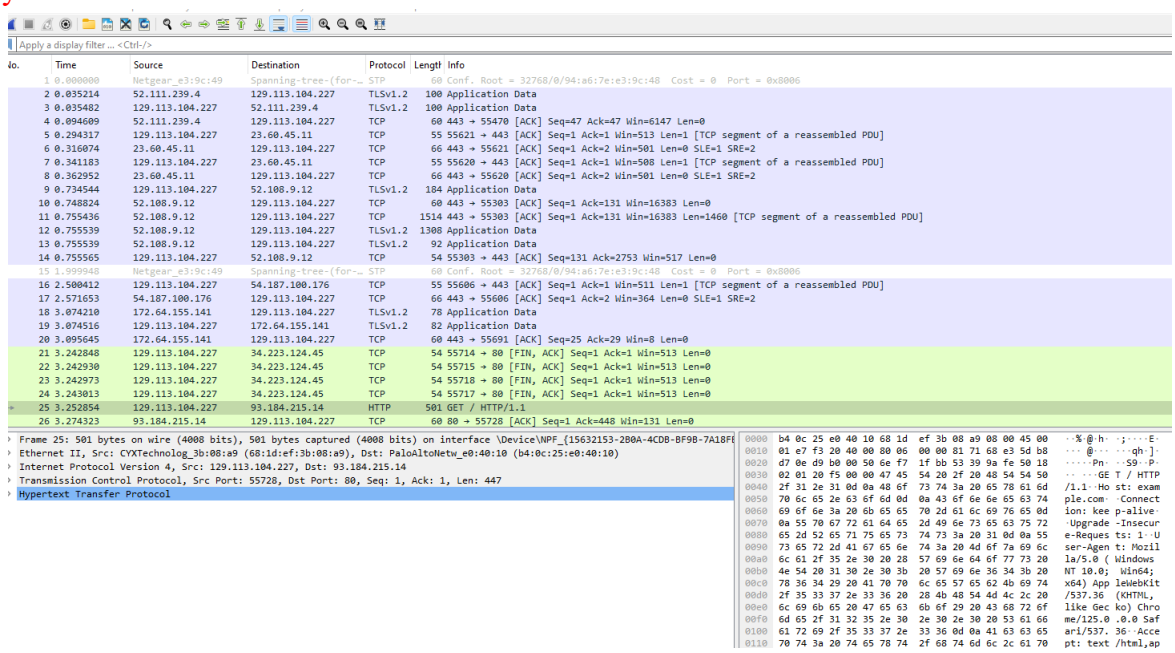


5. Capture some traffic from the site, and then hit c



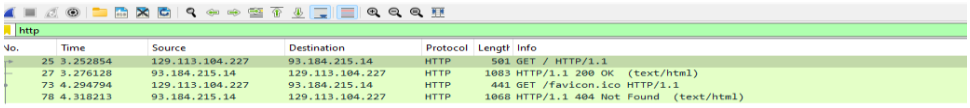
6. Your Wireshark should have packets numbered 1, 2, 3, on.

My wireshark:



7. Answer the following questions:

a. Type ***http*** in the filter area and hit enter (you can clear any filter by deleting the filter and hitting enter afterwards. If the http traffic is not supported, you might get one or two packets only with http and the rest with TLS traffic for the https supported site.



b. What is the IP address of the source and destination?

Source: 129.113.104.227

Destination: 93.184.215.14

c. Verify using nslookup the destination address of the school

```
Windows PowerShell

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\christopher> nslookup 93.184.215.14
Server:  pe-infns3.ad.utrgv.edu
Address:  129.113.38.34

*** pe-infns3.ad.utrgv.edu can't find 93.184.215.14: Non-existent domain
PS C:\Users\christopher>
```

d. Double click on the first http packet with the GET in the info and show the 5 layers

```
> Frame 25: 501 bytes on wire (4008 bits), 501 bytes captured (4008 bits) on interface \Device\NPF_{15632153-2B0A-4CDB-BF9B-7A18FE
> Ethernet II, Src: CYXTechnolog_3b:08:a9 (68:1d:ef:3b:08:a9), Dst: PaloAltoNetw_e0:40:10 (b4:0c:25:e0:40:10)
> Internet Protocol Version 4, Src: 129.113.104.227, Dst: 93.184.215.14
> Transmission Control Protocol, Src Port: 55728, Dst Port: 80, Seq: 1, Ack: 1, Len: 447
> Hypertext Transfer Protocol
```

e. For the packet above, expand the hypertext Transfer protocol layer (the application layer) and show what ***request*** was sent.

```
> Frame 25: 501 bytes on wire (4008 bits), 501 bytes captured (4008 bits) on interface \Device\NPF_{15632153-2B0A-4CDB-BF9B-7A18FE
> Ethernet II, Src: CYXTechnolog_3b:08:a9 (68:1d:ef:3b:08:a9), Dst: PaloAltoNetw_e0:40:10 (b4:0c:25:e0:40:10)
> Internet Protocol Version 4, Src: 129.113.104.227, Dst: 93.184.215.14
> Transmission Control Protocol, Src Port: 55728, Dst Port: 80, Seq: 1, Ack: 1, Len: 447
v Hypertext Transfer Protocol
  > GET / HTTP/1.1\r\n
    Host: example.com\r\n
    Connection: keep-alive\r\n
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/125.0.0.0 Safari/537.36\r
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-e
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: en-US,en;q=0.9,es-US;q=0.8,es;q=0.7\r\n
    \r\n
    [Full request URI: http://example.com/]
    [HTTP request 1/2]
    [Response in frame: 27]
    [Next request in frame: 73]
```

f. For the packet above, expand the Transmission Control Protocol layer (the Transport layer) and show what ports it has.

```
Source Port: 55728
Destination Port: 80
```

g. For the packet above, expand the Internet protocol layer (the Network layer) and show what IPS it has.

```
Internet Protocol Version 4, Src: 129.113.104.227, Dst: 93.184.215.14
```

h. For the packet above, expand the Ethernet II (the Link layer) and show what MAC Addresses it has.

```
> Destination: PaloAltoNetw_e0:40:10 (b4:0c:25:e0:40:10)
> Source: CYXTechnolog_3b:08:a9 (68:1d:ef:3b:08:a9)
  Type: IPv4 (0x0800)
```

8. Save your Wireshark trace (use file à save as) as LastnameFirstname_id. Submit this document and the trace together. You will receive a 00 if the trace is not submitted.